

# **SIMATS ENGINEERING**

## **ASSESSMENT TOOL 2**

**COURSE NAME: CSA07 COURSE CODE: Computer Networks**

---

### **COURSE OUTCOME COVERED**

**CO1:** *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

*Assessment Tool Weightage: 30%*

---

Annexure A

### **CONCEPT MAPPING**

---

#### ***Code of Conduct:***

*I SANDHIYA S (192521224) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

*I understand that any violation of academic integrity rules will result in disciplinary action.*

#### ***Signature of the student:***

SANDHIYA S

## Annexure A

### CONCEPT MAPPING

1. A partially completed concept map is given below:

Application → Presentation → \_\_\_\_\_ → Transport → Network → \_\_\_\_\_ → Physical.

Analyze the missing layers and explain how their functions are essential for reliable communication.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

3. Given the concept map:

DataLink → MACAddress → LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

4. A concept map shows:

ErrorControl

→ DataLinkLayer → ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

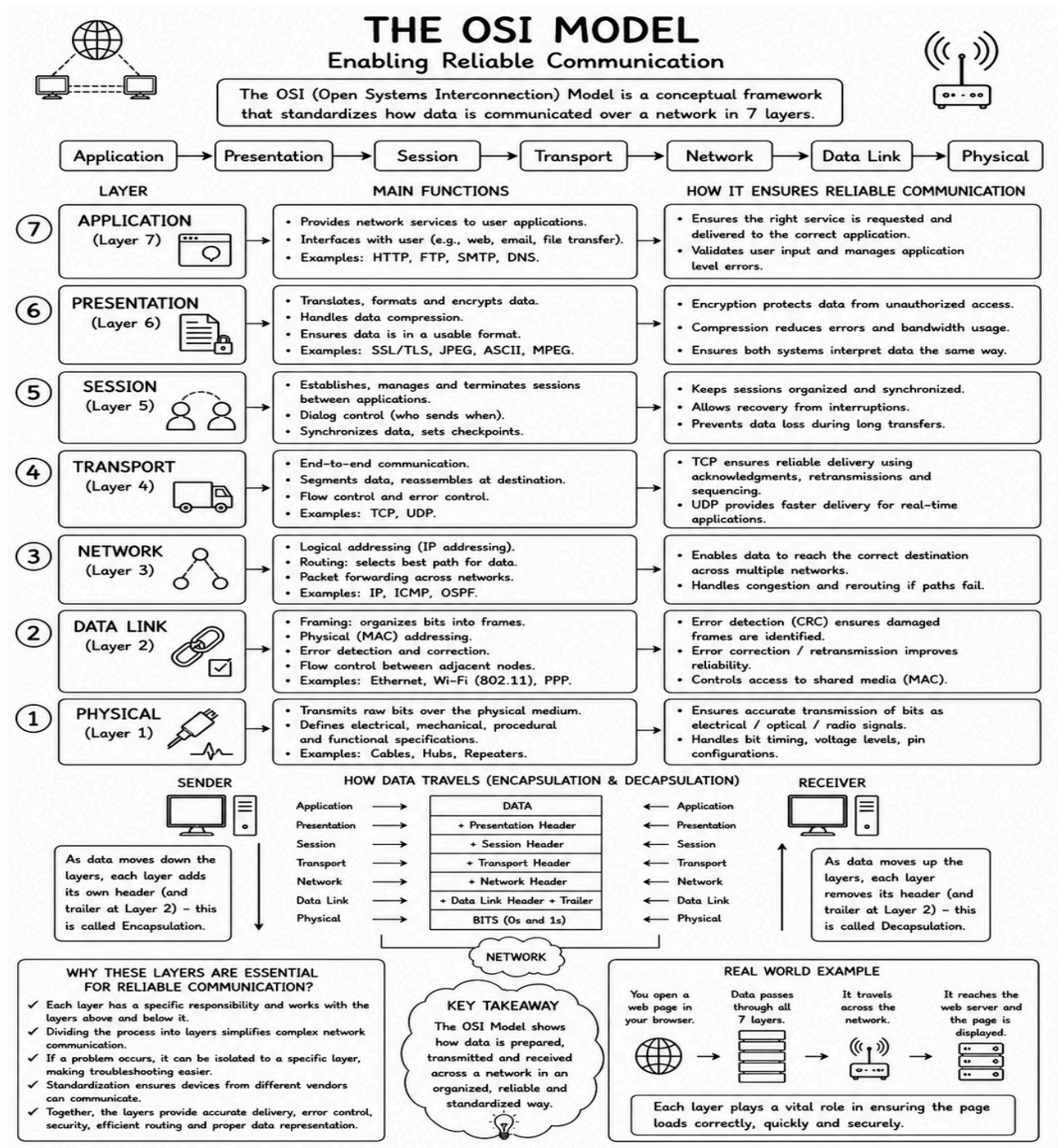
6. A network issue is represented in the concept map:

Physical ✓ → Data Link ✓ → Network ✗ → Transport ✗ → Application ✗

Analyze the problem and explain how the concept map helps in troubleshooting.

# ANSWERS

1.



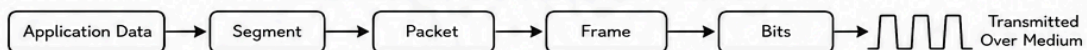


# OSI MODEL: DATA UNITS AND THEIR ROLE IN COMMUNICATION

Different layers of the OSI model use specific data units to encapsulate, control, and deliver data reliably from sender to receiver.

| OSI LAYER                                                                                 | DATA UNIT                         | CONCEPT MAP (ENCAPSULATION FLOW) | ROLE IN TRANSMISSION                                                                                                                                                       |
|-------------------------------------------------------------------------------------------|-----------------------------------|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>7 APPLICATION</b><br>Provides network services to end users (e.g., web, email)         | —<br>(No data unit at this layer) | Application Data (User Data)     | <ul style="list-style-type: none"> <li>User data is created and requested here.</li> <li>No data unit added.</li> </ul>                                                    |
| <b>6 PRESENTATION</b><br>Formats, encrypts, compresses data                               | —<br>(No data unit at this layer) | Application Data (User Data)     | <ul style="list-style-type: none"> <li>Ensures data is in a usable format (e.g., encryption, compression).</li> </ul>                                                      |
| <b>5 SESSION</b><br>Establishes, manages and terminates sessions                          | —<br>(No data unit at this layer) | Application Data (User Data)     | <ul style="list-style-type: none"> <li>Manages session and checkpoints.</li> </ul>                                                                                         |
| <b>4 TRANSPORT</b><br>End-to-end communication (reliability, flow control, error control) | <b>SEGMENT</b><br>(Data Unit)     | Segment                          | <ul style="list-style-type: none"> <li>Breaks data into segments, adds port numbers and sequence numbers.</li> <li>Ensures reliable delivery (e.g., TCP).</li> </ul>       |
| <b>3 NETWORK</b><br>Logical addressing and routing across networks                        | <b>PACKET</b><br>(Data Unit)      | Segment → Packet                 | <ul style="list-style-type: none"> <li>Adds logical addresses (IP addresses) and routing information.</li> <li>Part of packet travels across multiple networks.</li> </ul> |
| <b>2 DATA LINK</b><br>Reliable delivery on a single link (framing, error detection)       | <b>FRAME</b><br>(Data Unit)       | Segment → Packet → Frame         | <ul style="list-style-type: none"> <li>Adds MAC addresses and error detection (FCS).</li> <li>Ensures reliable delivery over a single link.</li> </ul>                     |
| <b>1 PHYSICAL</b><br>Transmits raw bits over a physical medium                            | <b>BITS</b><br>(Data Unit)        | Segment → Packet → Frame → Bits  | <ul style="list-style-type: none"> <li>Converts bits into electrical/optical/radio signals and transmits over the medium.</li> </ul>                                       |

## ENCAPSULATION FLOW (FROM SENDER TO RECEIVER)



## HOW DATA UNIT TRANSFORMATION SUPPORTS RELIABLE COMMUNICATION



- **Layered Encapsulation:** Each layer adds control information to the data, creating a data unit suited for its function. This ensures proper addressing, error checking, and sequencing.



- **Error Detection and Control:** Frame (Layer 2) and Segment (Layer 4) include mechanisms to detect errors and request retransmission if needed, ensuring data arrives correctly.



- **Addressing and Routing:** Packet (Layer 3) contains logical addresses that allow data to travel across different networks to its destination.



- **Segmentation and Reassembly:** Segment (Layer 4) divides large data into manageable pieces and reassembles them at the destination, improving efficiency and reliability.

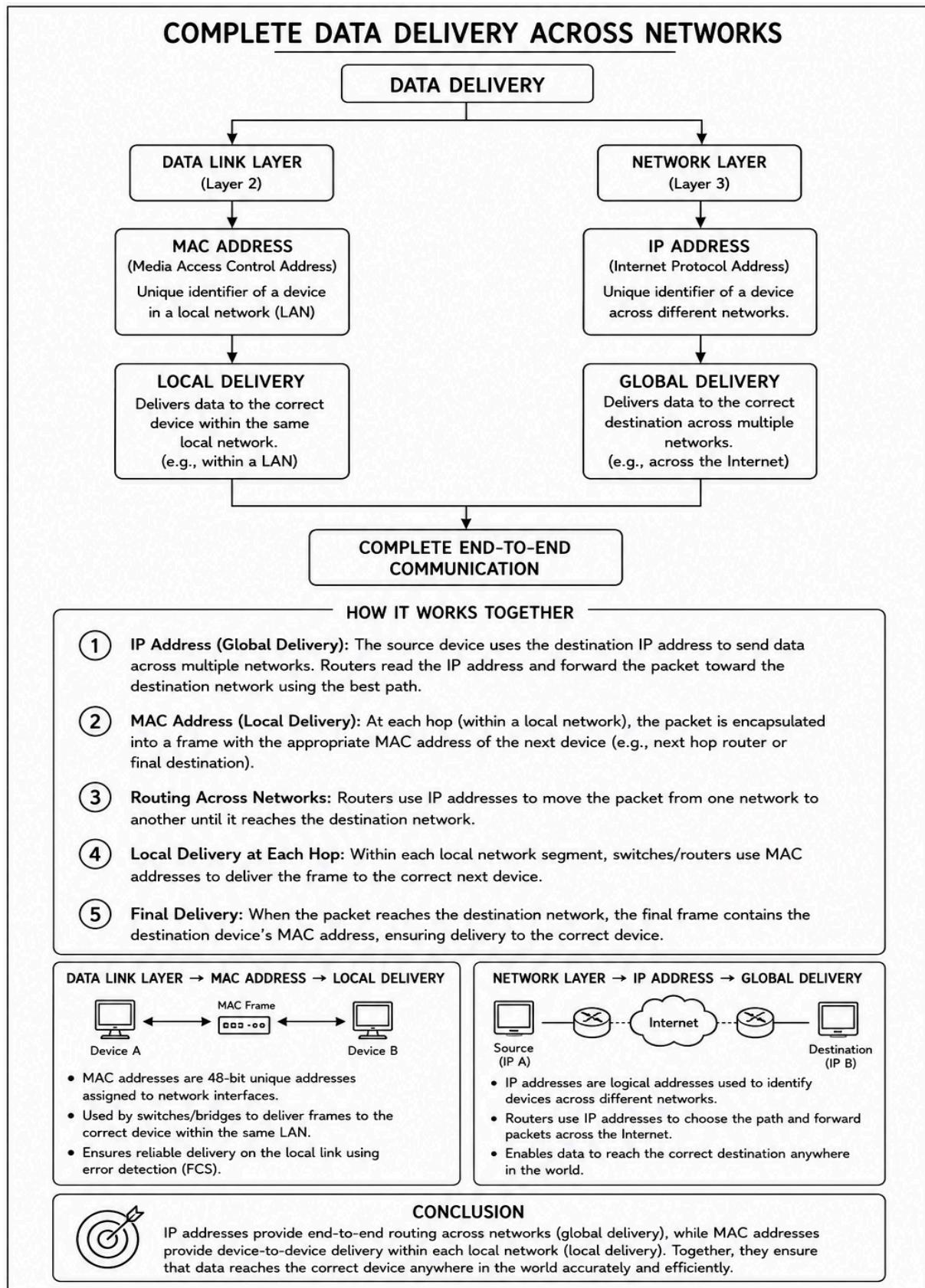


- **Physical Transmission:** Bits (Layer 1) convert data into signals that can travel over physical media, ensuring that data is physically delivered across the communication channel.

Together, these data units and layer functions provide a structured, reliable, and efficient way to send data from one device to another across any network.

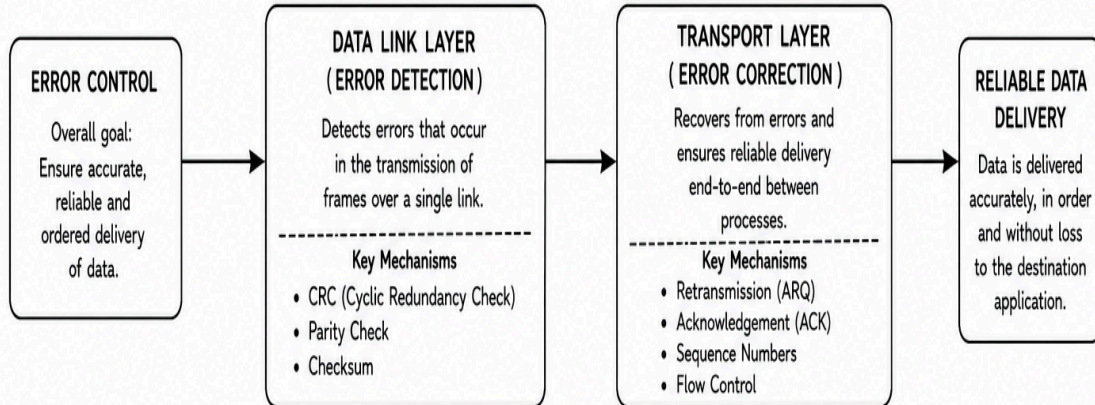


3.

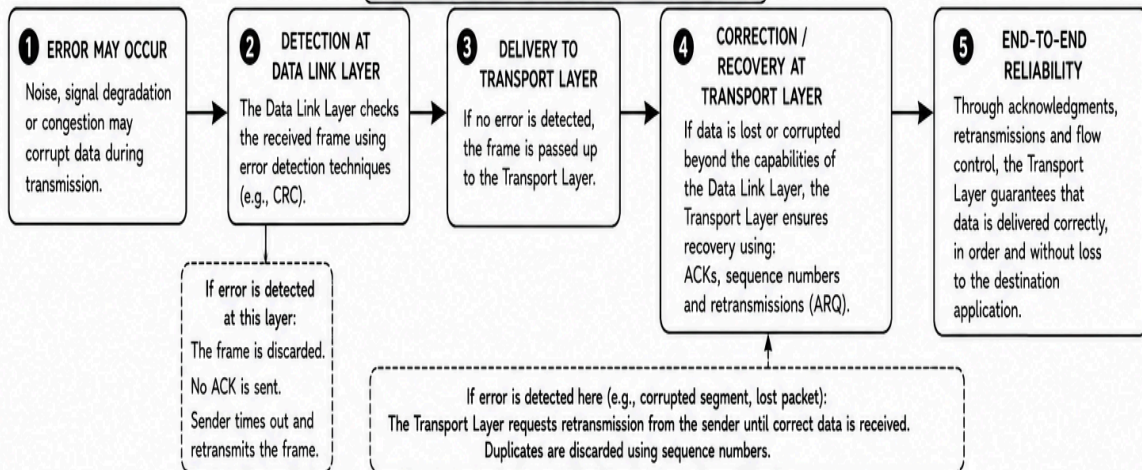


4.

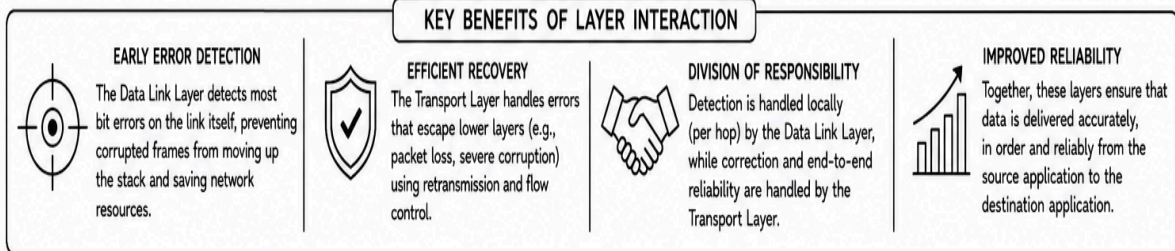
## HOW LAYERS WORK TOGETHER FOR RELIABLE DATA TRANSMISSION



### HOW THE INTERACTION IMPROVES RELIABILITY



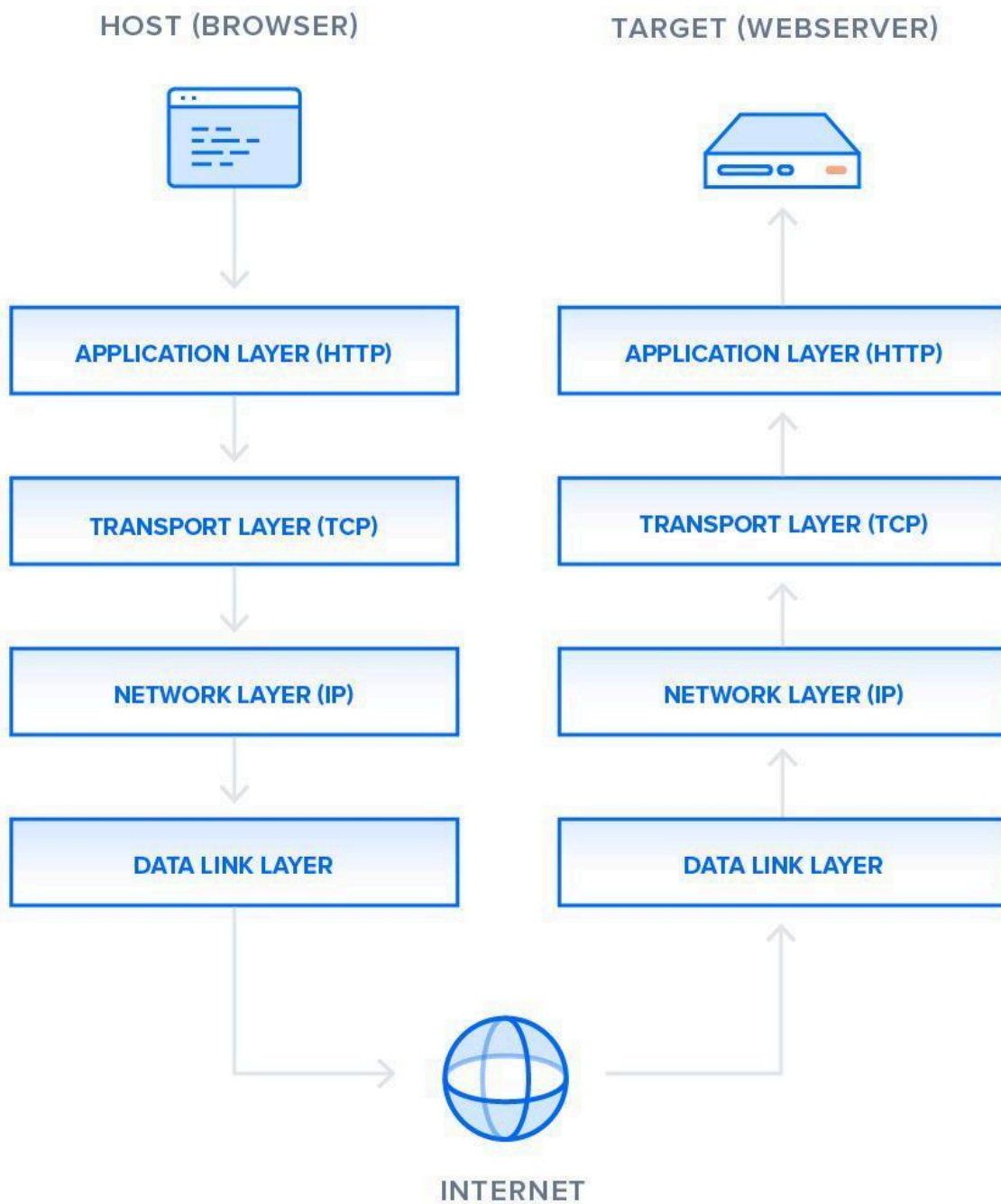
### KEY BENEFITS OF LAYER INTERACTION



**SUMMARY:** The Data Link Layer focuses on detecting errors in frames and preventing corrupted data from moving forward. The Transport Layer focuses on correcting errors that affect delivery and ensures end-to-end reliability. Their interaction provides a robust, efficient and reliable communication system.



5.





## NETWORK TROUBLESHOOTING ANALYSIS

### LAYER STATUS ANALYSIS

|   | OSI LAYER                                                                                  | STATUS                                                                                               | OBSERVATION                                                                                                                                                                                        |
|---|--------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ① | <b>PHYSICAL LAYER</b><br>Transmission of raw bits over physical medium                     | <br><b>WORKING</b>  | <ul style="list-style-type: none"> <li>Cables, switches, Wi-Fi, connectors are working.</li> <li>Physical connection established.</li> </ul>                                                       |
| ② | <b>DATA LINK LAYER</b><br>Framing, MAC addressing, error detection                         | <br><b>WORKING</b>  | <ul style="list-style-type: none"> <li>Frames are transmitted successfully.</li> <li>MAC addressing and error detection are working.</li> </ul>                                                    |
| ③ | <b>NETWORK LAYER</b><br>Logical addressing and routing (IP)                                | <br><b>FAILED</b> | <b>Failure occurs at this layer.</b> <ul style="list-style-type: none"> <li>IP addressing issue</li> <li>Default gateway issue</li> <li>Routing table error</li> <li>Router may be down</li> </ul> |
| ④ | <b>TRANSPORT LAYER</b><br>End-to-end communication, reliability (TCP/UDP)                  | <br><b>FAILED</b> | <ul style="list-style-type: none"> <li>Cannot establish end-to-end communication.</li> <li>Segments cannot reach destination.</li> </ul>                                                           |
| ⑤ | <b>APPLICATION LAYER</b><br>Network services to applications (HTTP, DNS, Email, FTP, etc.) | <br><b>FAILED</b> | <ul style="list-style-type: none"> <li>Applications cannot access network resources.</li> <li>Services appear down to the user.</li> </ul>                                                         |

### HOW THE CONCEPT MAP HELPS IN TROUBLESHOOTING



#### 1. IDENTIFY THE FIRST POINT OF FAILURE

The first failed layer is the Network Layer.



#### 2. NARROW DOWN THE PROBLEM

Since Physical and Data Link layers are working, the issue is not with hardware or local communication.



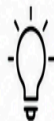
#### 3. SAVE TIME

Focus directly on Network Layer configuration instead of checking lower layers.



#### 4. FOCUS ON POSSIBLE CAUSES

- Check IP address configuration
- Verify subnet mask
- Check default gateway
- Verify routing table
- Check router status



#### 5. UNDERSTAND LAYER DEPENDENCY

Failure in the Network Layer prevents the Transport and Application layers from functioning.

### EFFECT OF FAILURE



### CONCLUSION

The concept map shows that the Network Layer is the root cause of the problem. Since OSI layers are interdependent, failure at the Network Layer prevents the Transport and Application layers from operating, causing complete communication failure.

### KEY TAKEAWAY

By identifying the first failed layer using the concept map, network administrators can troubleshoot efficiently, fix the root cause, and restore full communication.

## RUBRICS FOR CALCULATION

| Criteria                           | Weightage (%) | Level 4 (Exemplary)                                                                                          | Level 3 (Proficient)                                     | Level 2 (Developing)                                    | Level 1 (Beginning)                       |
|------------------------------------|---------------|--------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|---------------------------------------------------------|-------------------------------------------|
| <b>Concept Identification</b>      | 20%           | Accurately identifies all relevant OSI layers, concepts, and relationships                                   | Identifies most concepts with minor omissions            | Identifies limited concepts; some inaccuracies          | Unable to identify key concepts correctly |
| <b>Concept Mapping Structure</b>   | 25%           | Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections            | Concept map is mostly clear with minor structural issues | Concept map lacks clarity, organization, or proper flow | No clear structure or incorrect mapping   |
| <b>Linking &amp; Relationships</b> | 20%           | Clearly establishes correct relationships between layers, functions, and processes                           | Relationships are mostly correct with minor errors       | Limited or partially correct relationships              | Incorrect or missing relationships        |
| <b>Application &amp; Analysis</b>  | 20%           | Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning | The application is correct but lacks depth in analysis   | Basic application with weak explanation                 | Unable to apply concepts correctly        |
| <b>Explanation &amp; Clarity</b>   | 15%           | Provides a clear, concise, and wellstructured explanation supporting the concept map                         | Explanation is understandable with minor gaps            | Explanation is unclear or incomplete                    | No clear or meaningful explanation        |